

Spectrophotometer vs. spectroradiometer

Two instruments that both produce a spectrum, separated by where the light comes from and what the numbers mean.

In short: a spectrophotometer makes its own light and measures how a sample interacts with it. A spectroradiometer measures whatever light arrives, in absolute physical units.

Spectrophotometer

- **Has an internal light source** — tungsten-halogen, deuterium, or xenon flash — plus a sample holder in a fixed, controlled geometry.
- **Measures a ratio, not absolute energy:** transmittance or reflectance relative to a reference (a blank cuvette, a white Spectralon standard) measured through the same optical path. Output is unitless, reported as 0–1 or a percentage, or as absorbance $A = -\log_{10} T$.
- Because source, path length, and geometry are fixed, the ratio is stable and reproducible. That is the entire point of the design.
- **Typical instruments:** Cary 5000, Shimadzu UV-3600, and the integrating-sphere leaf clips such as the ASD or PSR contact probe and the LI-COR 6800 leaf spectrometer. Anything using an integrating sphere for leaf reflectance and transmittance is effectively operating in spectrophotometer mode.

Spectroradiometer

- **No internal source** — it is a passive detector pointed at a scene, calibrated to report absolute radiometric quantities: spectral radiance in $\text{W m}^{-2} \text{sr}^{-1} \text{nm}^{-1}$, or irradiance in $\text{W m}^{-2} \text{nm}^{-1}$.
- Requires a radiometric calibration traceable to a standard lamp, together with a defined field of view or a cosine-corrected diffuser.
- To obtain reflectance you take two measurements — target radiance and a white reference panel under the same illumination — and form the ratio yourself. Illumination is the sun, or whatever is there, so it changes between scans.
- **Typical instruments:** ASD FieldSpec, Spectra Vista SVC HR-1024i, PSR+, and the sensors aboard airborne and satellite imaging spectrometers such as AVIRIS-NG, EMIT, and PRISMA.

Side by side

	Spectrophotometer	Spectroradiometer
Illumination	Internal, constant, controlled	Ambient — sun, sky, or whatever is present
Native output	A ratio: reflectance, transmittance, absorbance	Absolute radiance or irradiance in physical units
Getting reflectance	Direct, in one step against a blank or white standard	Two steps: target radiance ratioed to a white panel
Scale of measurement	Leaf or sample	Leaf through canopy to whole scene
Dominant noise sources	Instrument drift, stray light, sphere substitution error	Sun angle, cloud, BRDF, atmosphere, background

Why it matters for hyperspectral trait work

For partial least squares regression models predicting nitrogen, LMA, chlorophyll, carotenoids, V_{cmax} , J_{max} , cellulose, and lignin, leaf-level spectra from a contact probe or integrating sphere — spectrophotometer-style, with a fixed artificial source — are far cleaner than anything measured passively. That is the main reason leaf-level trait models validate so much better than canopy-level ones.

When the same model is transferred to canopy spectroradiometer data, the degradation you observe is mostly illumination geometry, structural scattering, and background contribution. It is rarely a failure of the underlying chemistry–spectra relationship.

A note on terminology

Field instruments such as the ASD FieldSpec are spectroradiometers. Attach a contact probe with its own lamp, however, and you are using one as a spectrophotometer. The hardware category and the measurement mode are not the same thing, and papers are not always careful about the distinction.